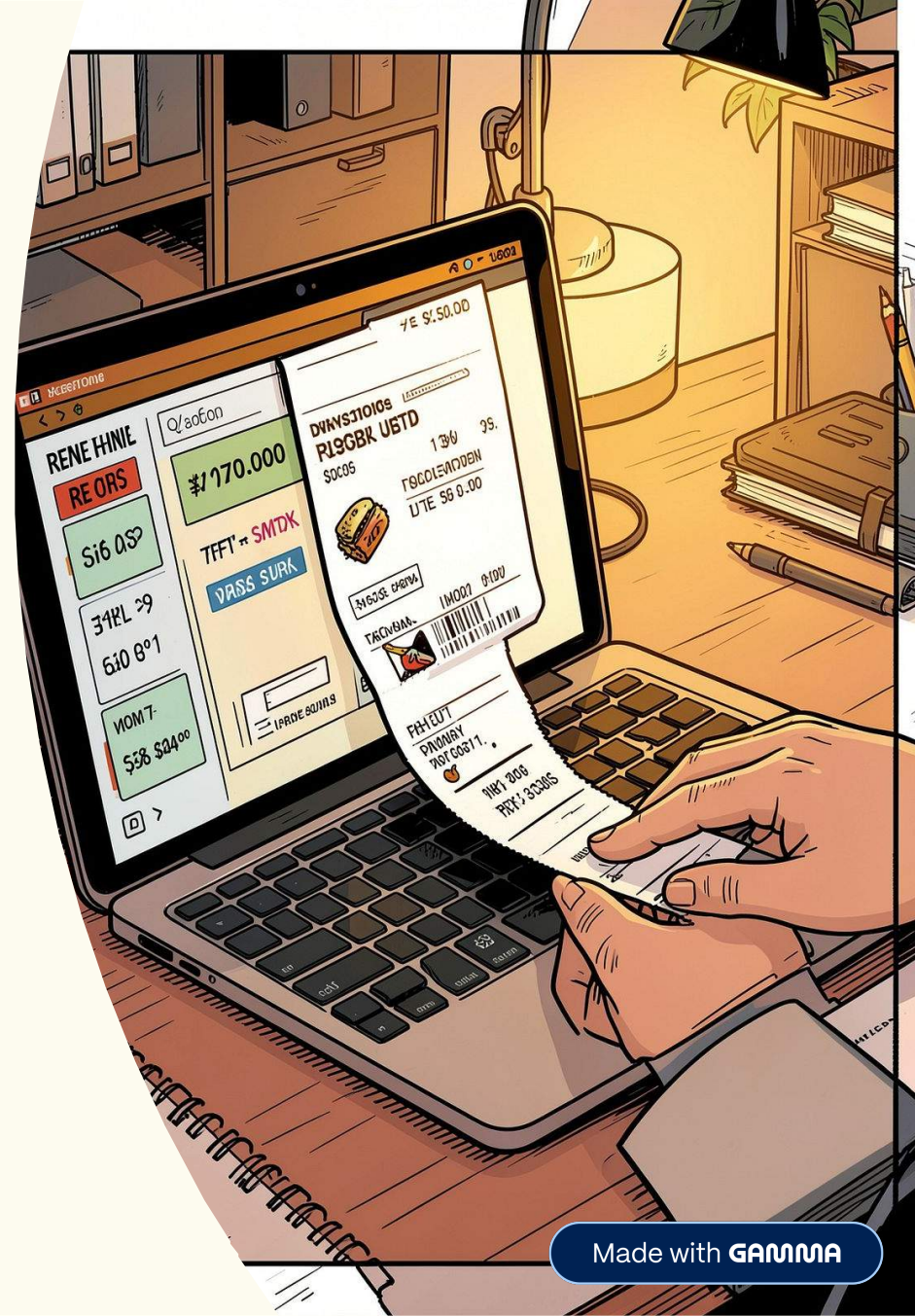


Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: uncover spending, segments, preferences, subscriptions



Dataset Summary

Rows & Columns

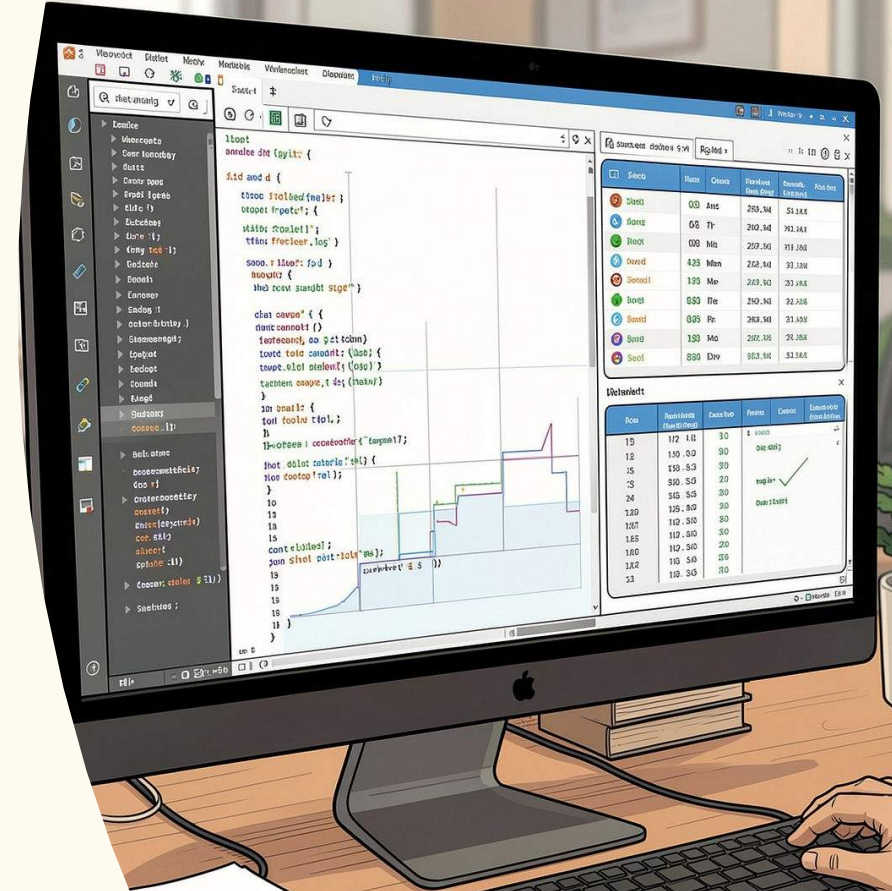
Rows: 3,900 •
Columns: 18

Key Features

Demographics,
Purchase details,
Behavior flags

Missing Data

37 values in
Review Rating





EDA — Python Workflow

01

Load & Explore

pandas: df.info(), .describe()

02

Clean & Impute

Imputed Review Rating by category median

03

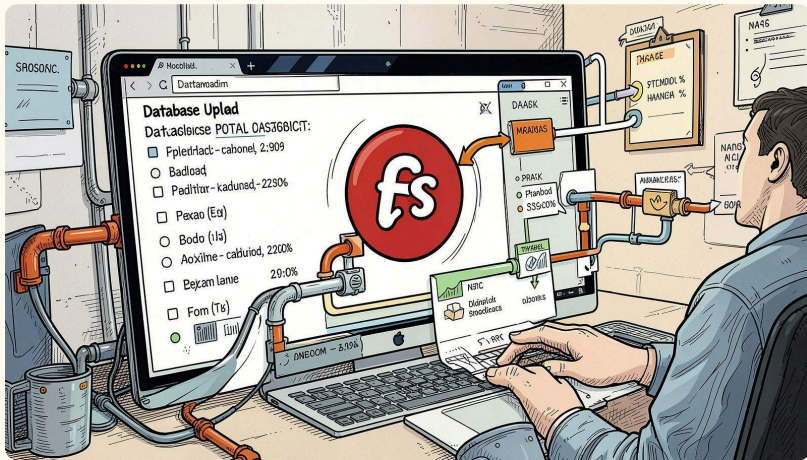
Feature Engineering

age_group, purchase_frequency_days

04

Standardize

Renamed columns to snake_case



Database Integration & Checks

Connected cleaned DataFrame to PostgreSQL

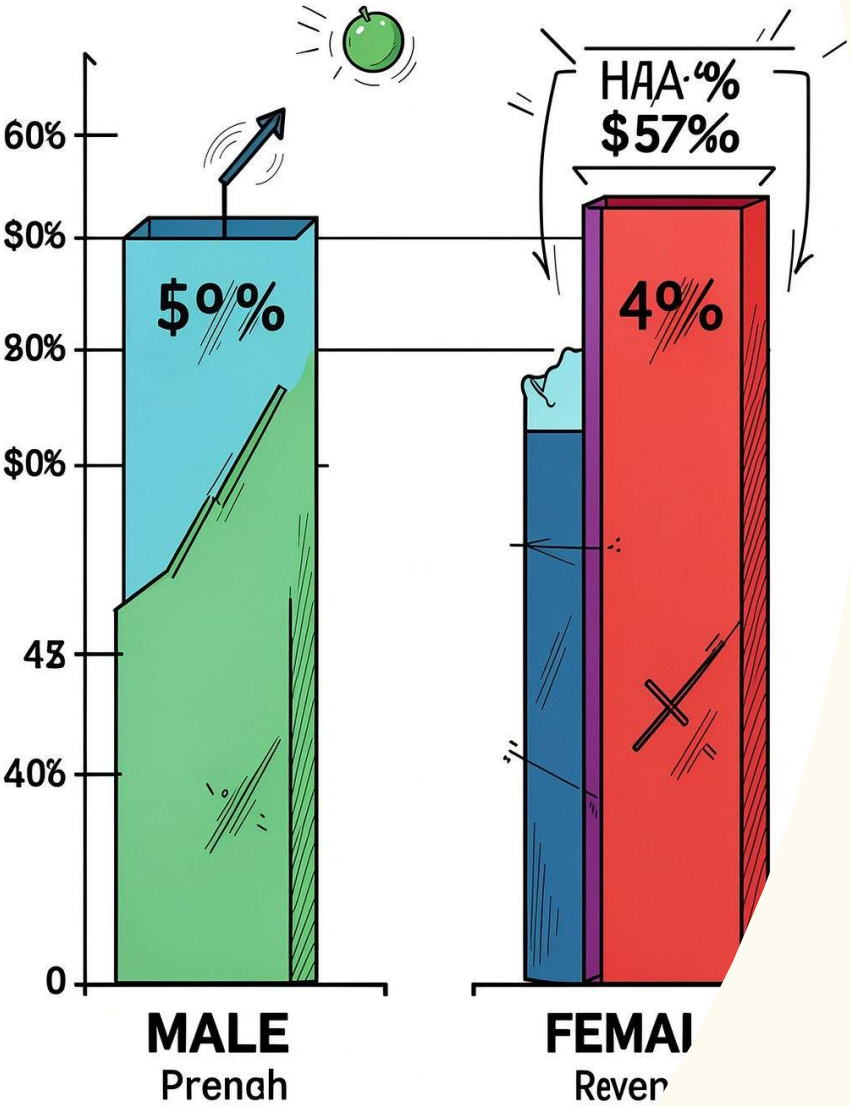
Consistency

Dropped redundant
`promo_code_used`
(redundant with
`discount_applied`)

SQL Ready

Prepared for business
queries and reporting

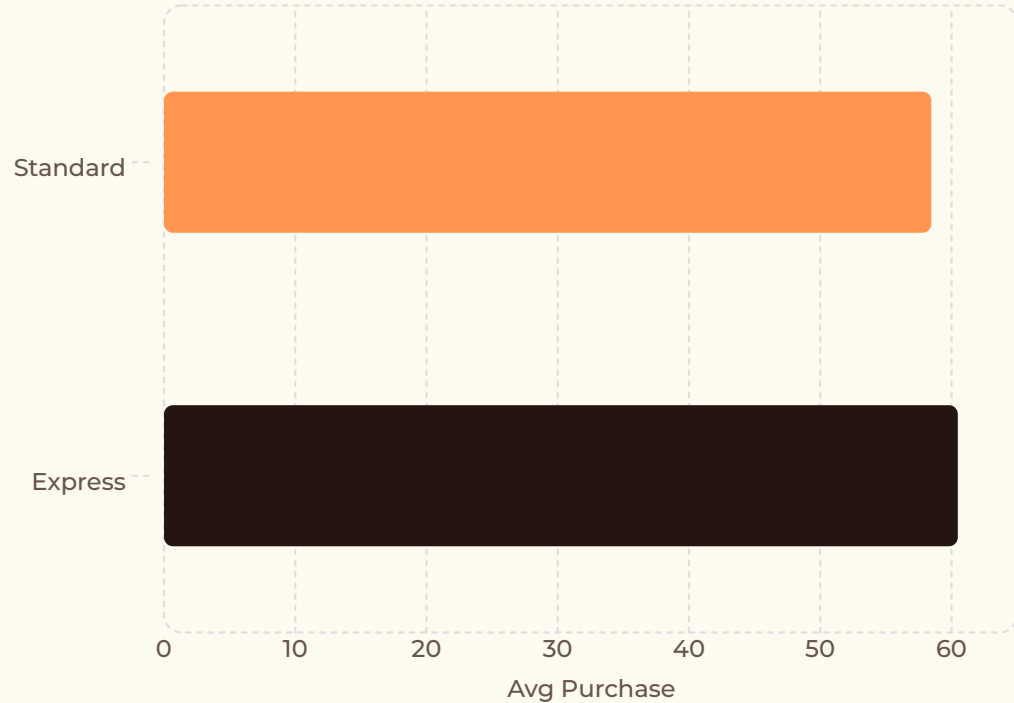
SHNRIE RNIVENUE



Revenue by Gender

- Male
\$157,890
- Female
\$75,191

Shipping Type



Shipping Comparison

Express slightly higher avg purchase

📌 Use express-shipping users for targeted offers

Discounts & High-Spenders



High-Spending Discount Users

839 rows — customers who used discounts yet spent above average



Top Discount-Dependent Items

Hat 50% • Sneakers 49.66% • Coat 49.07% •
Sweater 48.17% • Pants 47.37%

Top Products & Segments



Top Rated

Gloves 3.86 • Sandals 3.84 • Boots 3.82 •
Hat 3.80 • Skirt 3.78



Top by Orders

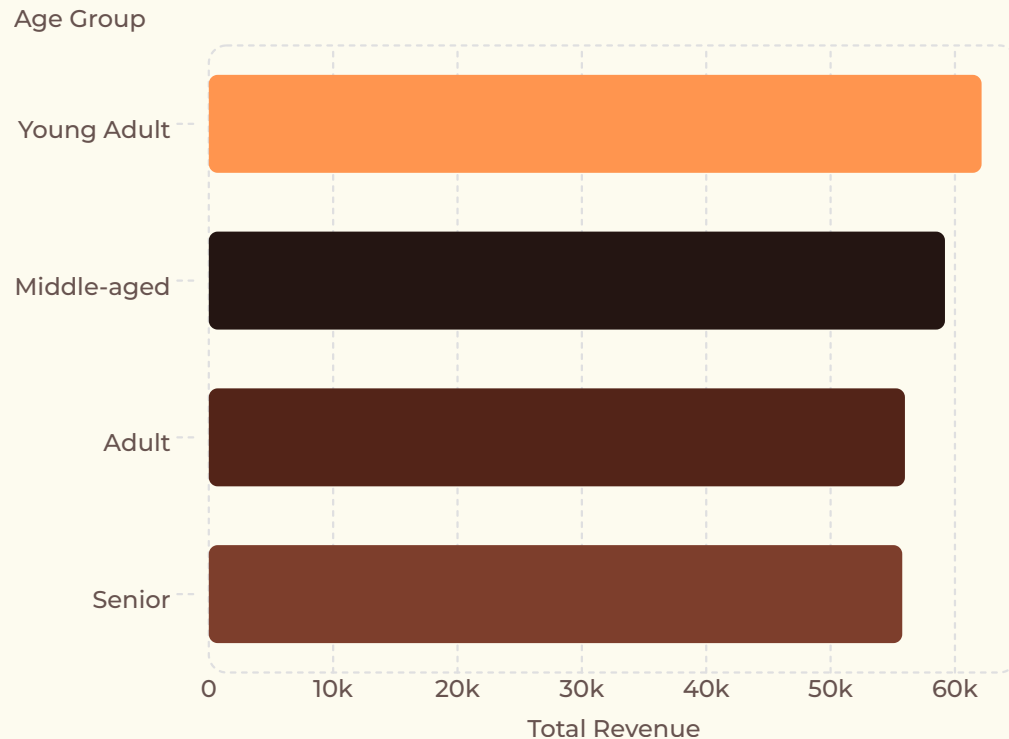
Jewelry, Sunglasses, Belt, Blouse,
Pants, Shirt



Customer Segments

Loyal 3,116 • Returning 701 • New 83





Revenue by Age Group

Focus marketing on Young Adult & Middle-aged segments

📌 Young Adult highest contributor: \$62,143

Recommendations & Next Steps

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discount Policy

Balance sales uplift with margins

Product Positioning

Feature top-rated & best-selling items in campaigns

Targeted Marketing

Prioritize high-revenue age groups and express-shipping users